BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 021 · LANs, WANs and Network Performance

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why is a thermostat controller an embedded system?

Q2. After one ascending bubble-sort pass on [5, 2, 4, 1], what is the list?
A. [2, 4, 1, 5]
B. [1, 2, 4, 5]
C. [5, 2, 4, 1]
D. [1, 4, 2, 5]

Q3. Insertion sort has sorted part [2, 5, 8] and next value 3. Which values move? Give the new sorted part.

Q4. A program tests whether score >= 50. “A comparison operator must always produce a text paragraph.” Correct this. What could happen?

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I Do · Read new knowledge

Key vocabulary

Network — Two or more connected devices sharing resources.

LAN — Local Area Network; small geographical area.

WAN — Wide Area Network; larger geographical area.

Bandwidth — Data a connection can transfer per second.

Transmission media — Wires, optical fibre or wireless signals carrying data.

New learning

Computer networks: Two or more connected computers share data and resources, such as printers, files and an internet connection.

LAN compared with WAN: A LAN covers a small area such as a school. A WAN covers a larger area and may link multiple LANs.

Network performance: Bandwidth measures how much data can be transferred per second. Users share available capacity; wired and wireless media can have different limits.

Explain the effect: Do not claim all WANs are slower. Use the scenario to link bandwidth, demand or transmission media to the impact on performance.

Worked example

A trust links two schools in different towns. Explain why it needs a WAN and how busy connections may affect performance. [4 marks]

1. Schools in different towns cover a larger geographical area.

2. A WAN links geographically distant networks [1] including the two school LANs [1].

3. Many simultaneous users share the available connection capacity [1], so file transfers may become slower [1].

4. Two WAN points and two linked performance points give four marking points.

We Do · Practise with guidance

Guided 1

A library connects its 15 PCs and a printer in one building. Explain why a LAN suits it and how many users could affect performance. [4 marks]

Worked method: A LAN covers a small area [1] and allows sharing the printer [1]. Many users share available bandwidth [1], which may slow transfers [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Guided 2

A school upgrades to a connection with more bandwidth. Explain two likely benefits. [4 marks]

Worked method: More data can be transferred per second [1], reducing transfer time for a file [1]. More simultaneous users may be supported [1] with fewer delays [1].

Guided 3

A classroom uses wireless tablets. Explain two factors that could slow the network. [4 marks]

Worked method: Many simultaneous users share bandwidth [1], slowing transfers [1]. Wireless interference [1] can disrupt data transmission and reduce reliability [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the estimated transfer time in seconds.

```
fileMb = 48
rateMbps = 12
print(fileMb DIV rateMbps)
```

predict

Question 1

State what LAN and WAN stand for. [2 marks] Source: Adapted from your Paper 1 revision booklet, section 1.3.1.

Question 2

An online gaming event at a youth club experiences delays when many players connect. Explain two factors affecting its network performance. [4 marks] Source: Original J277-style practice, not an OCR past-paper original.

Question 3

A retail company has a network within each shop and connects shops in different cities. Compare LANs and WANs and describe two factors that could affect performance. [6 marks] Source: Original J277-style practice, not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Every WAN is always slower than every LAN.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It is part of a heating system, with the specific task of controlling temperature.

Q2. A — [2, 4, 1, 5]

Q3. Values greater than 3 shift right: [5, 8]. New prefix [2, 3, 5, 8].

Q4. A comparison gives a Boolean result: true or false. It compares values; assignment stores a value.

I Do / We Do pages · Guided practice

Q1. A LAN covers a small area [1] and allows sharing the printer [1]. Many users share available bandwidth [1], which may slow transfers [1].

Q2. More data can be transferred per second [1], reducing transfer time for a file [1]. More simultaneous users may be supported [1] with fewer delays [1].

Q3. Many simultaneous users share bandwidth [1], slowing transfers [1]. Wireless interference [1] can disrupt data transmission and reduce reliability [1].

You Do pages · Code First

```
fileMb = 48
rateMbps = 12
print(fileMb DIV rateMbps)
```

Expected result: 4

You Do pages · Independent questions

Q1. Local Area Network [1]; Wide Area Network [1].

Q2. Many users share capacity [1], which may slow individual transfers [1]. Limited bandwidth [1] limits the amount of data transferred per second [1].

Q3. LAN small area [1], WAN large geographical area [1]. Bandwidth affects data transfer per second [1] and may change transfer speed [1]. Number of users affects available bandwidth per device [1], potentially causing delays [1].

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: